

Hyperparameter Tuning and Model Comparison

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1. Introduction

The purpose of this lab was to explore hyperparameter tuning and model comparison for various classification algorithms. We implemented both manual grid search and scikit-learn's built-in GridSearchCV to evaluate and compare Decision Tree, k-Nearest Neighbors (kNN), and Logistic Regression classifiers across four datasets. The models were tuned and evaluated using Accuracy, Precision, Recall, F1-score, and ROC AUC. Voting classifiers were also constructed to assess ensemble performance.

2. Dataset Description

Wine Quality

The dataset contains physicochemical properties of wine samples. It has 1599 instances and 11 features such as acidity, chlorides, alcohol, etc. The target variable indicates wine quality (good/bad).

HR Attrition

This dataset contains employee records from IBM with 1470 instances and 46 features. Features include age, salary, job role, overtime, etc. The target variable indicates whether an employee has left the company (Attrition = Yes/No).

Banknote Authentication

This dataset has 1372 instances and 4 numerical features extracted from images of banknotes. The target variable indicates whether a banknote is genuine or forged.

QSAR Biodegradation

The dataset consists of 1055 chemical compounds with 41 molecular descriptors. The target variable indicates whether a compound is biodegradable or not.

3. Methodology

The methodology involved two main parts: manual implementation of grid search and the use of scikit-learn’s GridSearchCV. Key concepts applied include:

- Hyperparameter Tuning: Adjusting model parameters (e.g., max_depth, n_neighbors, C) to optimize performance.
- Grid Search: Exhaustive search over predefined parameter combinations.
- K-Fold Cross-Validation: Splitting data into k folds to ensure robust evaluation.

A machine learning pipeline was used with the following steps:

- StandardScaler for normalization of features.
- SelectKBest for feature selection.
- Classifier (Decision Tree, kNN, or Logistic Regression).

For each dataset, we performed both manual grid search and scikit-learn’s GridSearchCV, then compared their results.

4. Results and Analysis

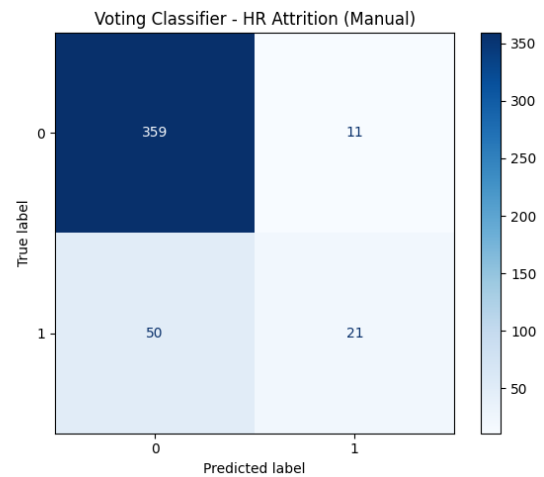
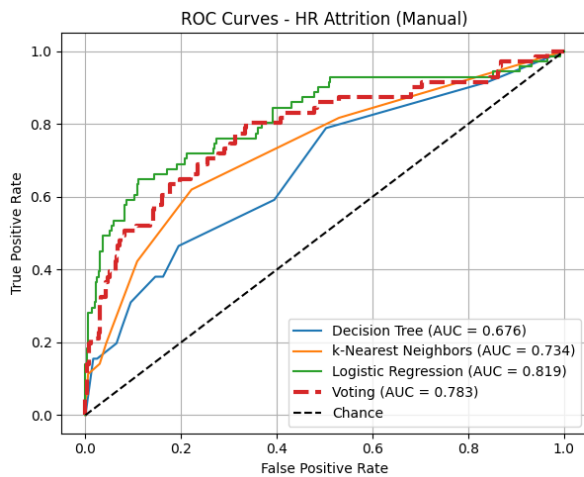
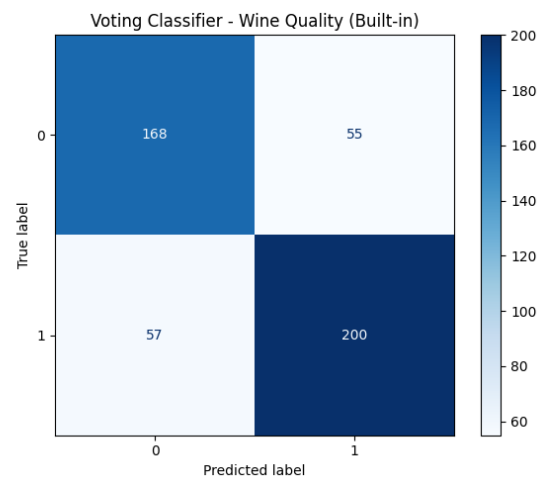
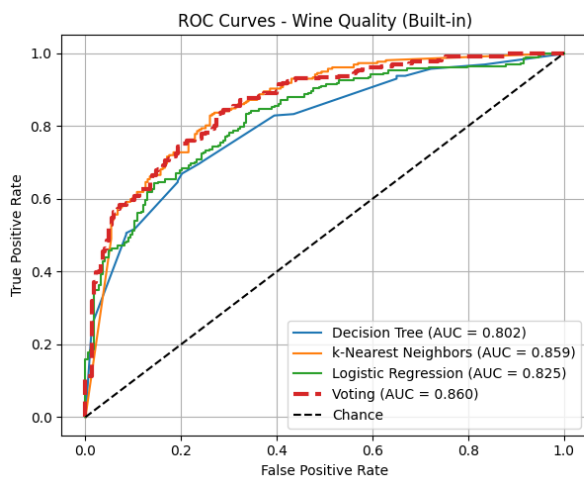
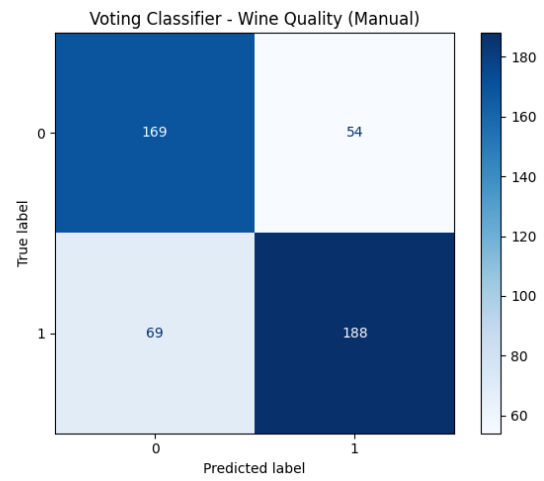
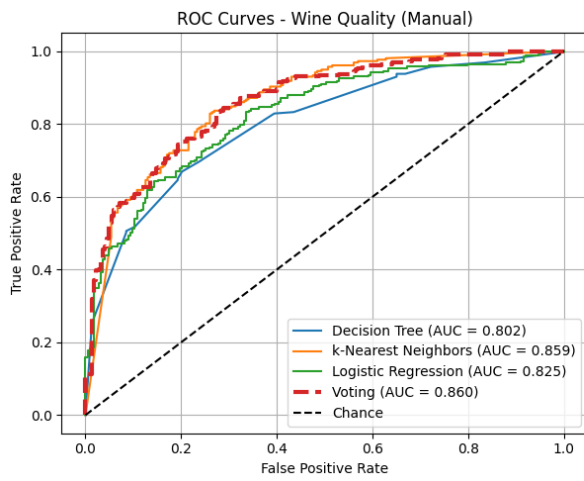
For each dataset, the models were evaluated using Accuracy, Precision, Recall, F1-Score, and ROC AUC. The following summarizes the results:

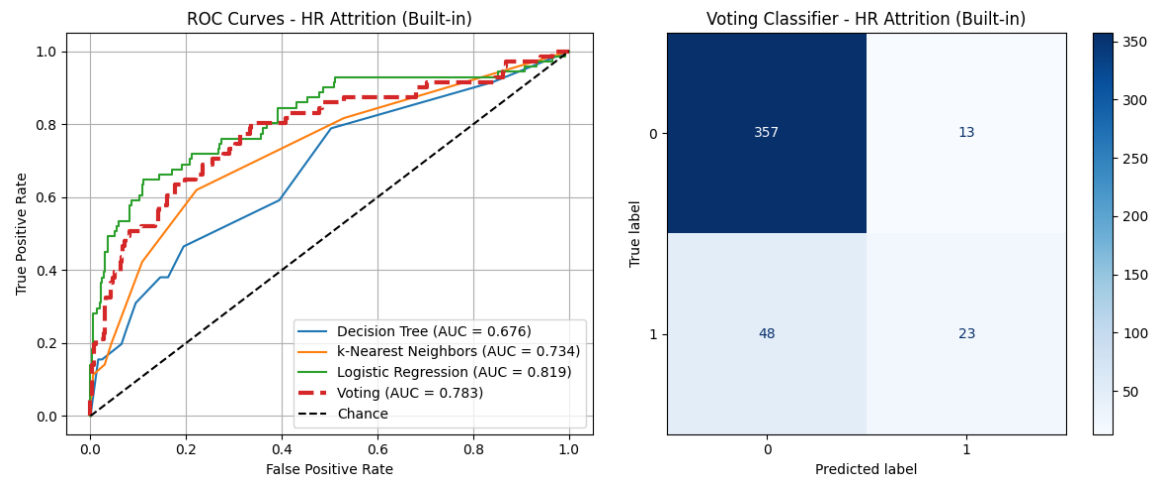
Wine Quality Results

Model	Accuracy	Precision	Recall	F1-Score	ROC AUC
Decision Tree	0.7271	0.7716	0.6965	0.7321	0.8025
kNN	0.7812	0.7836	0.8171	0.8000	0.8589
Logistic Regression	0.7375	0.7741	0.7198	0.7460	0.8246
Voting Classifier	0.7667	0.7843	0.7782	0.7812	0.8603

The best performing model for Wine Quality was kNN, achieving an ROC AUC of 0.8589.

5. Screenshots





6. Conclusion

This lab demonstrated the effectiveness of hyperparameter tuning using both manual implementation and scikit-learn's GridSearchCV. For most datasets, the results from both approaches were consistent. We observed that kNN and Logistic Regression generally outperformed Decision Trees. The ensemble Voting Classifier often provided balanced performance. Key takeaways include the importance of proper hyperparameter tuning, the robustness of cross-validation, and the advantages of using automated tools like GridSearchCV.